LCIS

Laboratory of Conception and Integration of Systems
Laboratoire de Conception et d'Intégration des Systèmes

2025-2026

Vincent Beroulle, Directeur

Ionela Prodan, Directrice adjointe

Agenda

- All began about 30 years ago
- Research environment
- Teams
- Resources - platforms
- Scientific animation
- New scientific challenges

Let’s go back about 30 years ago

- In **1996**, Grenoble INP decided to strengthen its presence in **Valence** by creating **Esisar** (École nationale supérieure en systèmes avancés et réseaux), a new engineering school dedicated to **embedded systems, automation, electronics and computer science**.

- At the same time, the **Laboratory of Conception and Integration of Systems (LCIS)** was founded as the **first research laboratory** on the Campus Valence. Its mission was to provide **scientific support** for Esisar’s teaching and to develop **cutting-edge research** in systems design and integration.

People involved in 1996

Co-funders of LCIS

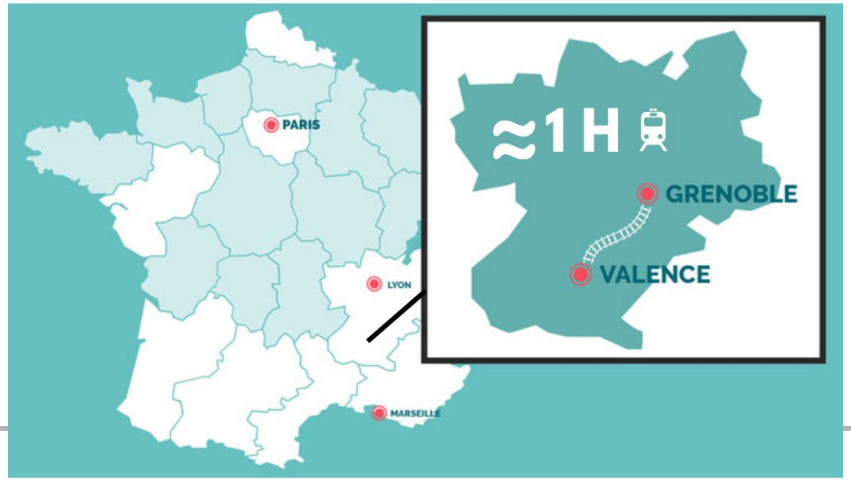
- **Smail Tedjini (Professor in RF)**
- **Denis Genon-Catalot (Assoc. Prof. in Telecommunication and Networks),**
- **Andre Lagreze (Assoc. Prof. in Telecommunication, now retired)**

Why was created the LCIS lab in Valence

- **To link research and education** - provide opportunities for lecturers and students from ESISAR school to engage in timely research and innovation.
- **To support regional talent** - offer projects and training opportunities to students in the Region
- **To collaborate with local industry** - build partnerships with the many companies (in technology, electronics and digital innovation) in this dynamic industrial area.

To recap

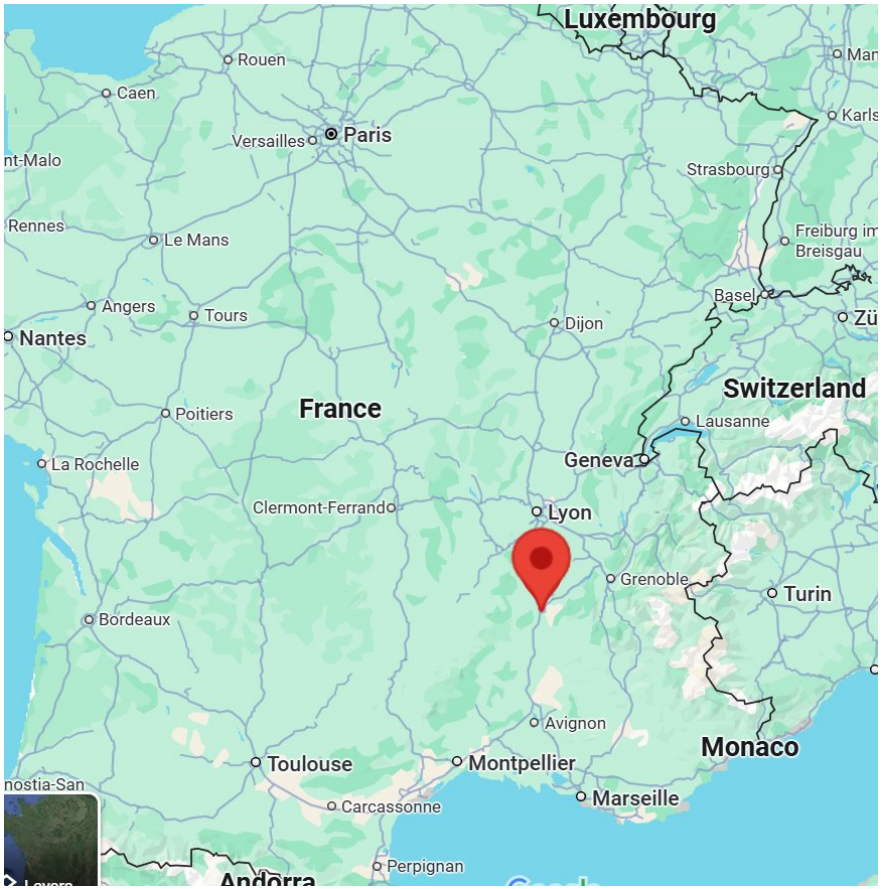
- Founded in **1996**
- **Hosted by Grenoble INP-UGA** with Esisar Engineering School in Valence
- The only research laboratory in Valence and in the Drôme department
- Hosts researchers from both **Grenoble INP-UGA and UGA**
- One entity of the **Campus UGA Valence**, the France biggest decentralized campus (more than 4000 students)



Grenoble INP - Univ. Grenoble Alpes

- Graduate schools of Engineering and Management
  - 8350 students,
  - 38 laboratories,
  - 465 partner companies
  - in the **Engineering and Technology** area,

  is ranked the 93rd position worldwide, the 5th position in France, i.e., **the 1st institution outside the Paris region.**



Research environment

- LCIS produces about **20 journal papers and 60 conference contributions per year**.
- This ensures **scientific visibility, fosters collaborations and strengthens funding and industry partnerships**, which are crucial for the lab's growth and impact.

Associated laboratories of the Grenoble INP - UGA schools

Mathématiques, Sciences et Technologies de l'Information-Communication - MSTIC	
GIPSA-lab	Images, parole, signal, automatique
G-SCOP	Conception, optimisation et production
LCIS	Conception et intégration des systèmes
LIG	Informatique ambiante et durable
LJK	Mathématiques appliquées - Informatique
TIMA	Techniques de l'informatique et de la microélectronique pour l'architecture des systèmes intégrés
TIMC-IMAG	Techniques de l'ingénierie médicale et de la complexité - Informatique, mathématiques et applications
VERIMAG	Systèmes embarqués

Physique des Particules, Astrophysique, Géosciences, Environnement et Ecologie – PAGE	
GIPSA-lab	Images, parole, signal, automatique
IGE	Géosciences de l'environnement
ISTerre	Sciences de la Terre
LEGI	Mécanique des fluides et transferts
LiPhy	Physique
LPSC	Physique subatomique et cosmologie

Sciences Sociales - PSS	
CERAG	Recherches appliquées à la gestion
GAEL	Economie appliquée

Sciences Humaines et Sociales - SHS	
GIPSA-lab	Images, parole, signal, automatique
LIG	Informatique ambiante et durable

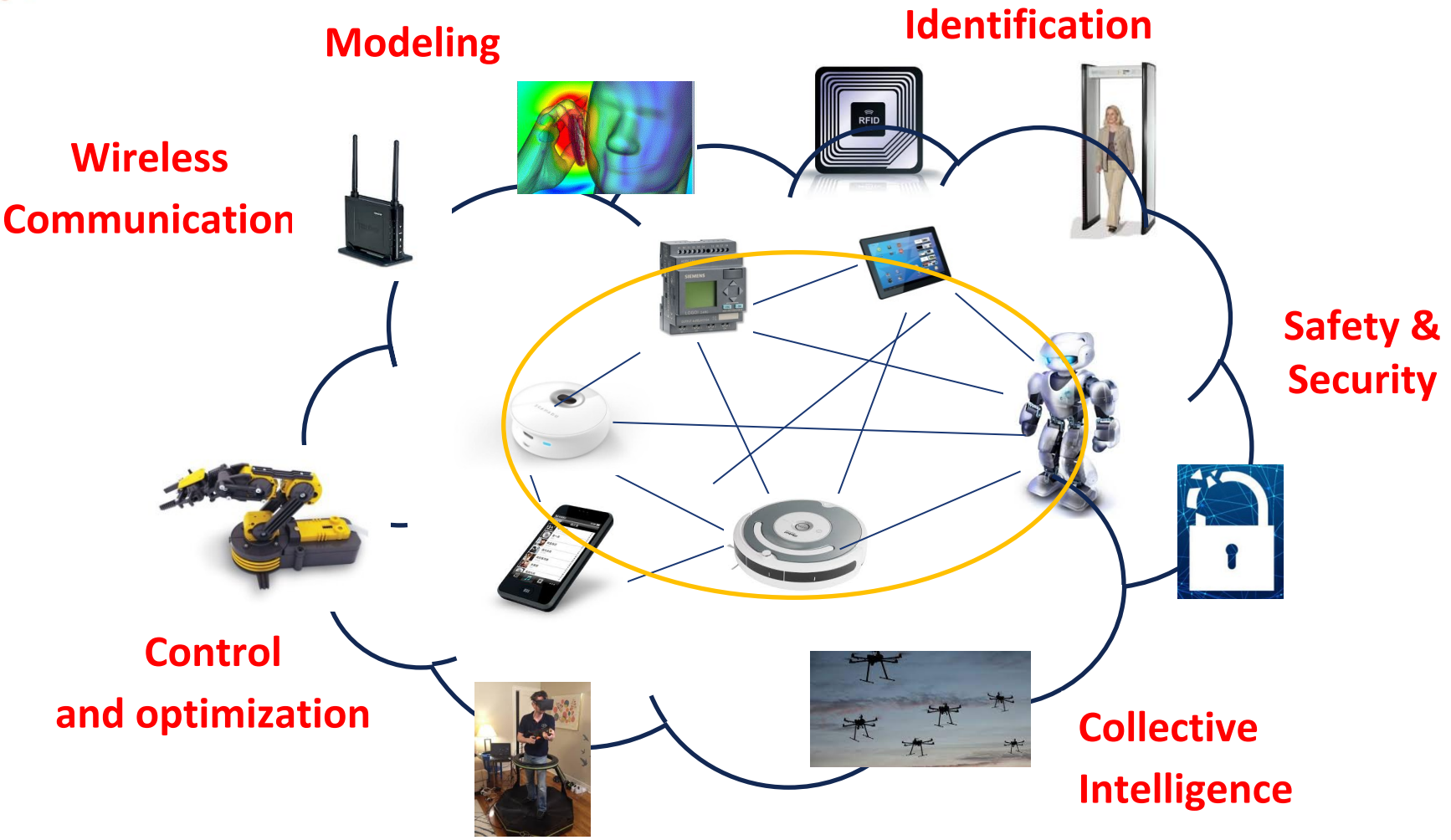
Physique Ingénierie et Matériaux - PEM	
3SR	Sols, solides, structures et risques
G2Elab	Génie électrique
G-SCOP	Conception, optimisation et production
IGE	Géosciences de l'environnement
CROMA (ex IMEP-LaHC)	Micro et nanoélectronique, microphotonique, micro et nanosystèmes, microondes et optomicroondes
LCIS	Conception et intégration des systèmes
LEGI	Mécanique des fluides et transferts
LEPMI	Electrochimie et physico-chimie des matériaux et des interfaces
LGP2	Génie des procédés papetiers
LiPhy	Physique
LMGP	Matériaux et génie physique
LRP	Rhéologie et procédés
LTM	Technologies de la microélectronique
NEEL	Recherche fondamentale en physique de la matière
Pheliqs	Photonique, électronique des matériaux
RFIC-Lab	Radio-fréquences et intégration de circuits
SIMaP	Science et ingénierie des matériaux et procédés
SPINTEC	Spintronique et technologie des composants
SyMMES	Systèmes moléculaires et nanomatériaux pour l'énergie et la santé
TIMA	Techniques de l'informatique et de la microélectronique pour l'architecture des systèmes intégrés
TIMC-IMAG	Techniques de l'ingénierie médicale et de la complexité - Informatique, mathématiques et applications

Scientific Field

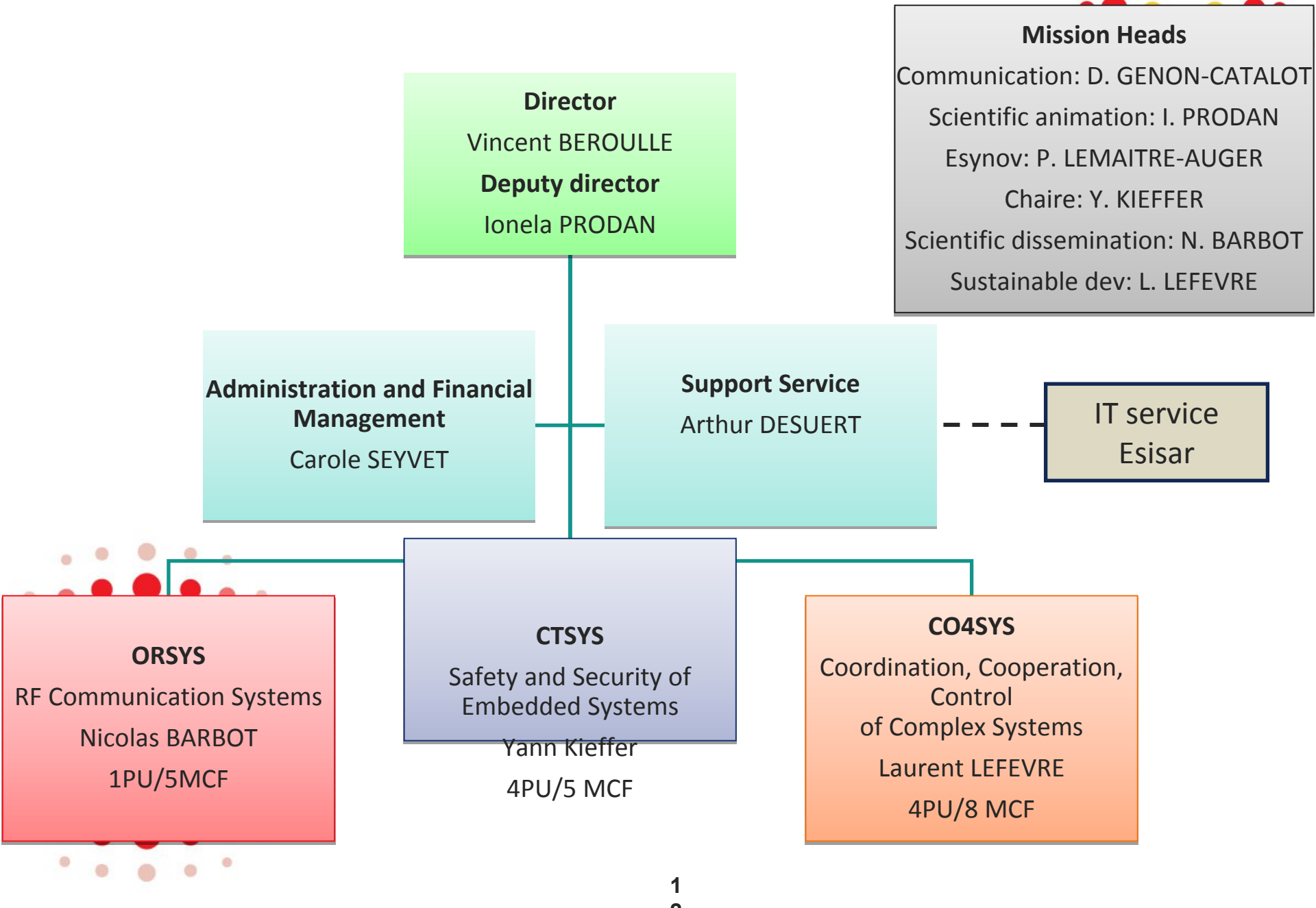
- **Cyber-Physical Systems (CPS)**
  - Development of methods, models, and tools for the design and the integration of **embedded and communicating systems** into physical environments

Embedded and Communicating Systems

Components, interactions, and usages



Structuration de la recherche



Research environment

Well integrated within the site's ecosystem

- **PEM** and **MSTIC** research clusters **(~13 projects)**
  - **CDP** (Cross Disciplinary Programs): Circular, CyberAlpes
  - Labex **PersyvalLab**, **Microelectronique/FMNT** Federation
  - **MIAI and Cybersecurity** Institutes, **IRT** Nanoelec
  - **Idex UGA**
- Two doctoral schools: **EEATS**, **MSTII**

- **Rhône-Alpes Region and International (~7 projects)**

- **Industrial transfer: IESA, Floralis, Institut Carnot, Linksium-SATT (~38 projects)**

- **Minalogic, Tenerrdis and Polymeris competitiveness clusters**

- **Chair ESSAIM-CPS (2025-2029) (2PhDs, Postdocs, internships, formation…)**
- **Chair TRUST (2015-2020) (2 PhDs + internships)**

PERSYVAL-Lab

Cybersecurity Institute
Univ. Grenoble Alpes

FMNT

chaire trust

NANOELEC.

MIAI Grenoble Alpes

La Région Auvergne-Rhône-Alpes

Research environment

- **Involved in various national projects (~9 projects)**
  - PEPR, ANR, ANRT, PGMO, DGA, BPI France

- **Involved in various European projects (~5 projects)**
  - Trust in HW/SW components, IoT security, energy management, etc.

- **Involved in mobility projects through which we have various invited Professors and Researchers** (CRCT, PHC, Erasmus+, Pack Ambition International, etc.) **(~8 projects)**

De nombreux partenariats industriels

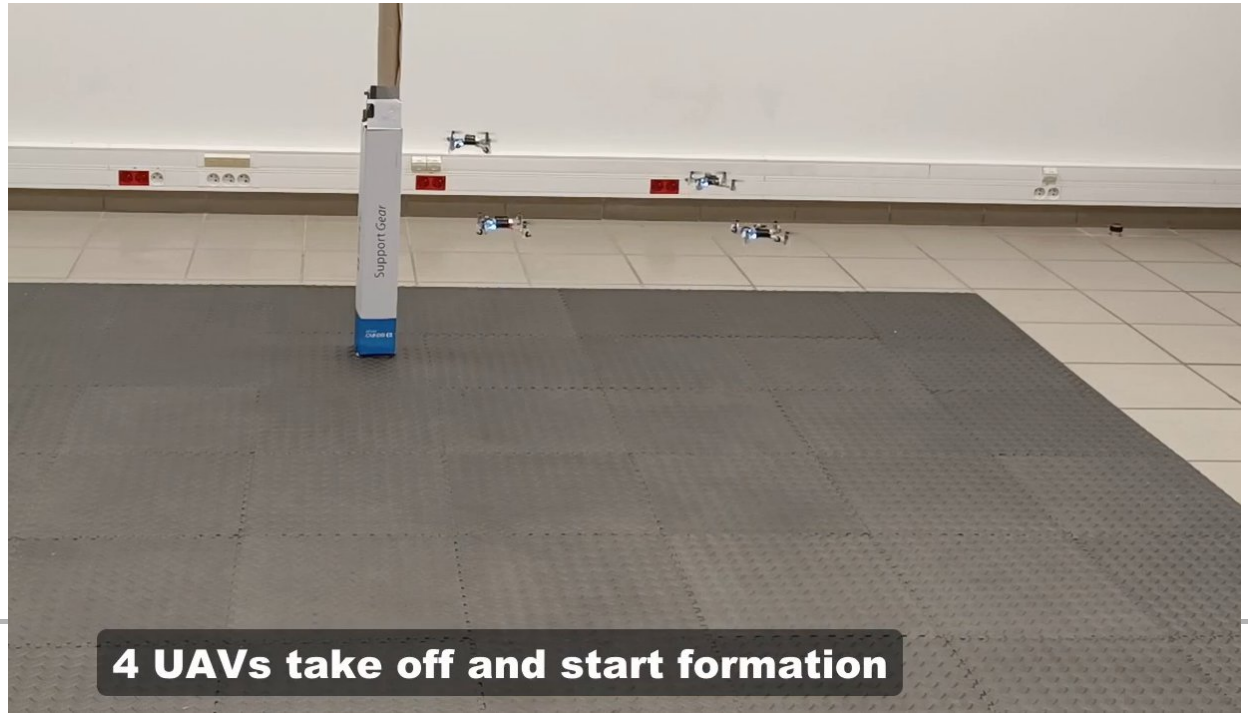
Examples of projects with companies

- **Control of a three-phase arc furnace** producing Si-Mg (Sillico-Manganese)
  - CIFRE PhD thesis of Minh Tuan Dinh
  - Eramet Ideas a company situated in Trappes, Paris area
  - The idea is to implement a model predictive controller

  on a real furnace in Dunkirk producing 60 000 tones of SI-Mg
  - the collaboration continues in a national ANR project, Control EAF

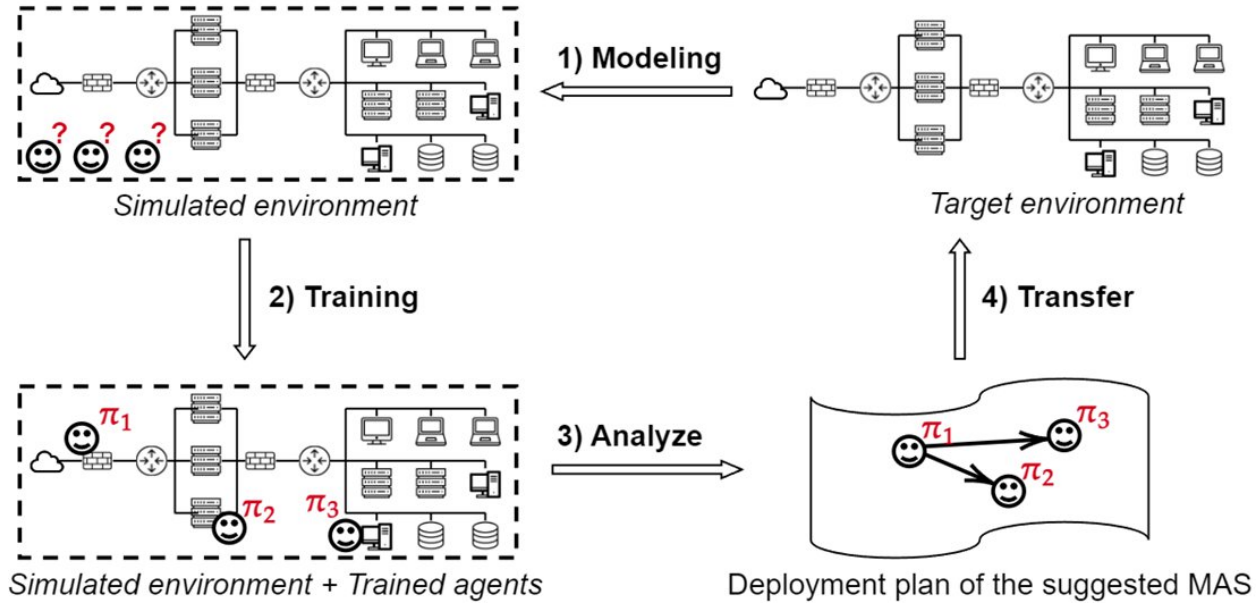
Examples of projects with companies

- PhD thesis of Vincent Marguet (student of Esisar, ISC specialisation)
- **Indoor formation motion** planning using B-splines parametrization and evolutionary optimization
- Vincent is now R&D Engineer at SurveyCopter



Examples of projects with companies

- PhD thesis of Julien Soule in collaboration with **Thales :** ``On the Organization of a Cyberdefense Multi-Agent System”



LCIS Teams

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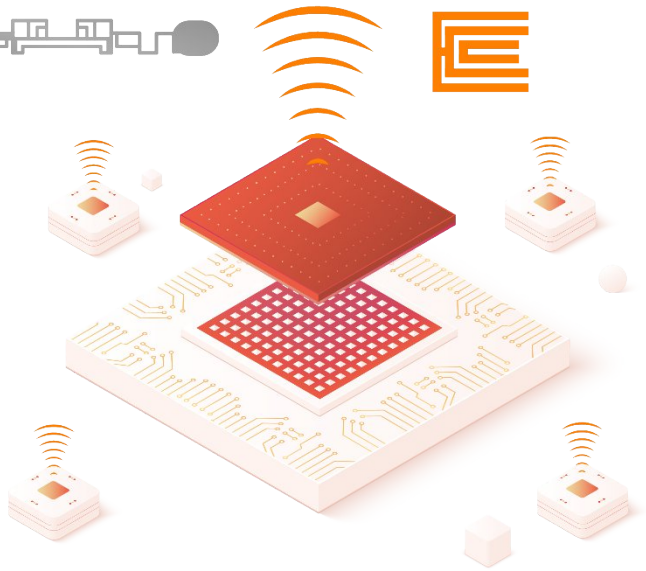
Laboratoire de Conception et d'Intégration des Systèmes



ORSYS

Radiofrequency Communication Systems

N. Barbot

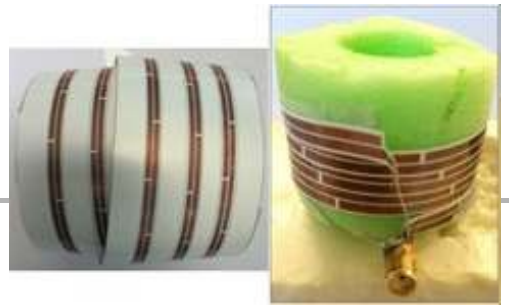
Staff Member

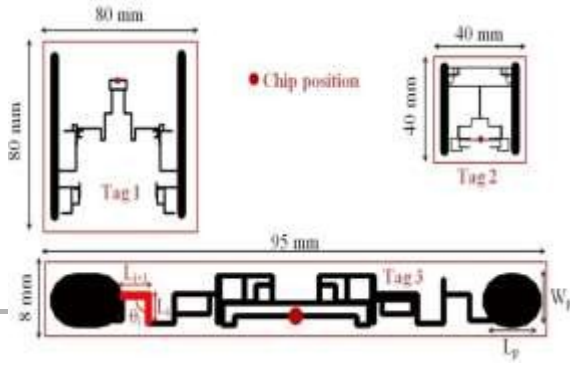
- 7 Permanents (1 Pr Emerite, 1 Pr, 5 MCF)
- MCF Nicolas Barbot
- MCF Darine Kaddour
- MCF Romain Siragusa
- Pr Etienne Perret
- MCF Pierre-Lemaitre Auger
- MCF Louis Morge-Rollet
- Pr. Emerite Smail Tedjini

- All members in the RF domain
- Small(er) team
- Multiples responsabilites

Scientific Issues and Research Areas

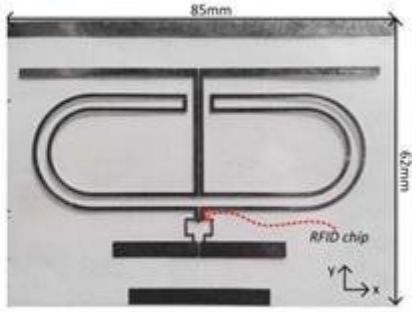
- Scientific issues
  - Development of wireless solutions and systems, from design to implementation on real systems
  - Characterization, measurement of physical quantities by RF approach
- Research areas
  - Specific antennas for communications
  - Passive transmission systems of information based on the principles of retro-modulation and retro-scattering of EM waves

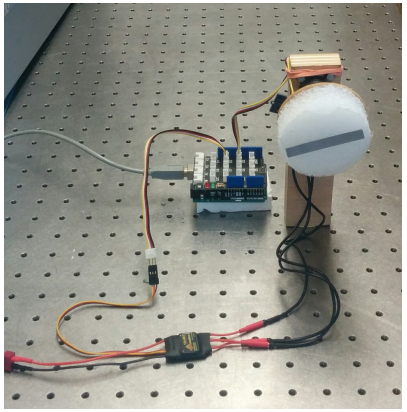




Main Application Areas

- Radio frequency identification
  - UHF RFID
  - Chipless RFID
- Specific antennas and circuits
  - Communications
  - Specific designs (wearable, food industry, MID, ...)





Main Results and Outcomes

- Chipless Technology
  - CBRAM for reprogrammable chipless tag
  - Gesture recognition
  - Sensing
- Linear Time-Variant and Non Linear transponders
  - Delta RCS
  - Optimal Matching
  - Polarization, Doppler, RCS modulations
  - Non-linear modulation
- Harmonic RFID
  - Proof of concept: Design of the harmonic reader

Undergoing Projects

- Australe (ANR), IMEP-LAHC, LCIS, CTP **Chipless Tag Authentication based on paper substrate in the mmWave band.** 1 PhD, 160k€
  Leader: F. Garet
- ScatterID (ERC), (LCIS, IES, ICCF, LTM) **Chipless Technology Developement and New Functionnalities**. 3 PhD, 9 years of post-PhD, 2M€
  Leader: E. Perret
- HRFID (Maturation Linksium), **Harmonic Signal Exploitation in UHF Technology**, 1 post-PhD, 200k€
  Leader: S. Tedjini
- Centrurion (Safran, DGA, INP), **Ergonomic Antenna Design integrated to military jacket,** 1 post-doc, 180k€
  Leader: S. Tedjini
- Wyzeflow (R&D Booster), **Motitoring of healthcare equipment using RFID technology**, 1 post doc, 110k€
  Leader N. Barbot

Scientific Production

Type	2014-2018	2018-2023
Journals	59	70
Conferences	85	72
National conferences	26	15
Books	7	2

- Constant production compared to the previous period
- 62 % of the journals are published in IEEE Transactions and journals
- Publication in national conferences to keep connexion with the community
- 10% of the production published with international colleagues

Collaborations

- International collaboration : University of Pisa, University of Tokyo, Herriot Watt University
- National collaboration : ICCM, IES, CEA Leti, LTM, TIMA
- Industrial collaboration : Dracula, Safran, Eolane, CTP, EDF, Michelin, Scimabio, DGA, MGI, Idyllic, Radial
- Involvement in scientific commitees: URSI, IEEE, JNM

Short Term Perspectives

- Scientific Perspectives:
  - RF link design on FPGA and associated security tools
  - Non Linear Transponders (ID and sensing)
  - Specific antennas, Arrays of Antennas
  - Wireless activation of CBRAMs
  - New functionalities in chipless RFID
  - RF components of innovative environments

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CTSYS

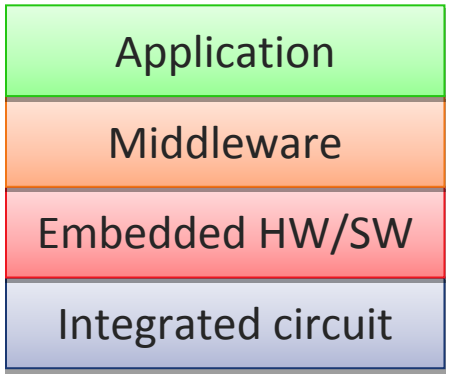
Safety and Security of Embedded and Distributed Systems

Yann Kieffer

Staff Member

- 9 Permanents (4 Pr, 5 MCF)
- Pr Laure Gonnord
- Pr Vincent Beroulle
- Pr Ioannis Parissis
- Pr Oum-El-Kheir Aktouf
- MCF Stephanie Chollet
- MCF Valentin Egglof
- MCF Christophe Deleuze
- MCF Nicolas Fourty
- MCF Sébastien Jean

Scientific topics and research areas

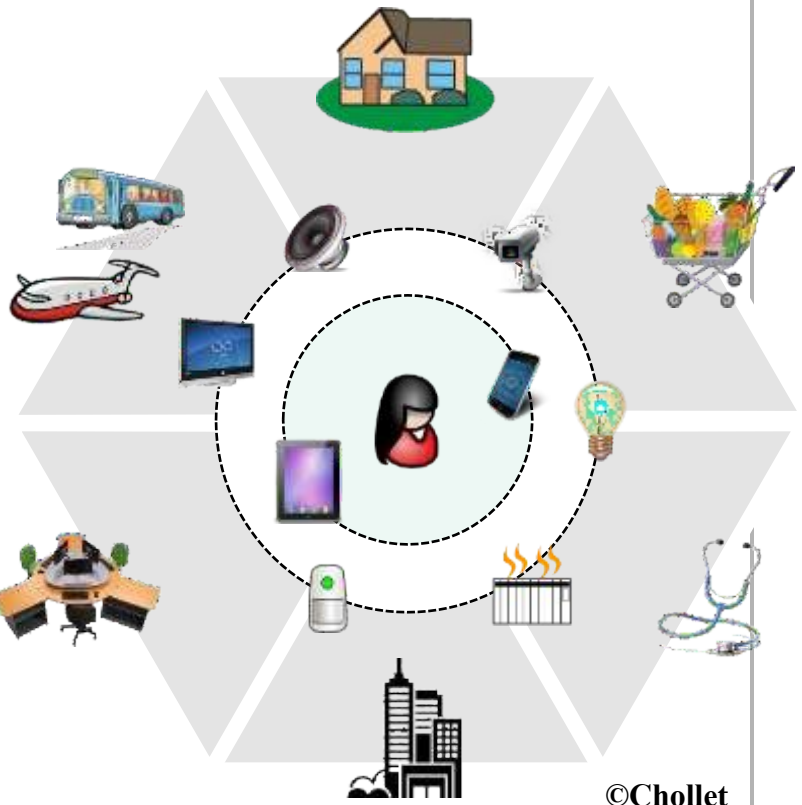


Topic 2: safety and security of distributed and pervasive applications and systems
(security analysis, security-by-design, decentralized security)

Topic 1: hardware and hardware/software systems security
(HW security, compilation-based security, security-oriented co-design)

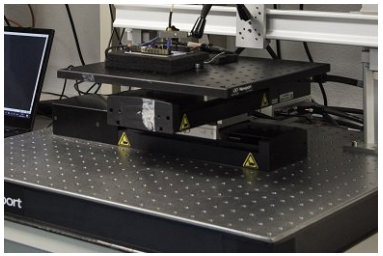
Main applications and societal challenges

- IoT and Connected Objects: RFID, UWB
  - Users security and privacy
  - Cost-effective security
- Critical Systems: avionics, medical objects
  - Users security, safety, and privacy
  - User trust
- Smart Systems: Smart Grids, Autonomous Vehicles
  - Users security, safety, and privacy
  - User trust
  - Energy sharing and saving



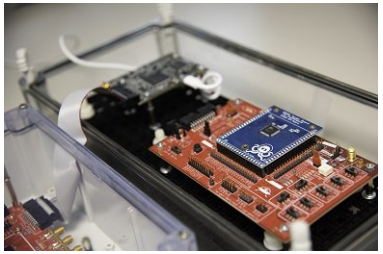
Experimental platforms and tools

- Sacco platform:
  - Hardware and embedded systems security tools
  - Shared with Esisar
- CyberNetX platform:
  - Indoor geolocation solutions based on UWB technology
  - Shared with Esisar, ORSYS and CO^{4}SYS



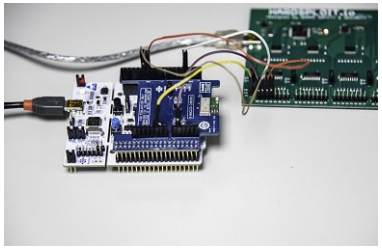
Electromagnetic fault injection unit

Side-channel analysis unit



Clock disruption attack

Climatic chamber



IoT security auditing

Secpump

Image credits: Esynov

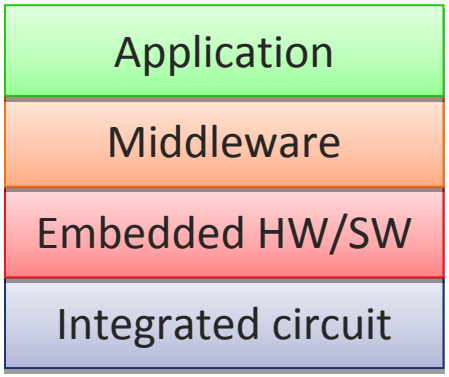
Human resources – Permanent members

- Distribution according to CNU sections:
  - 27 ☾ 70%
  - 61 ☾ 20%
  - 63 ☾ 10%

→ **Expertise complementarity**

- Presence within SOC2, GPL and Security GDRs

Main results and outcomes Portfolio

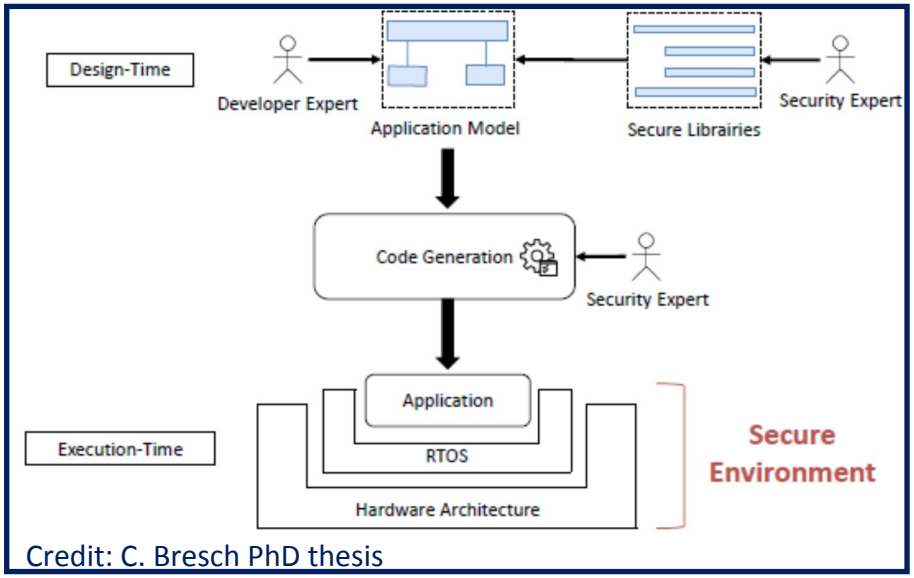


- A Survey on Parallelism and Determinism to analyze safety properties
- **A Practical Framework for Fine-Grained Control Flow Integrity in Critical Systems**
- Systematic Mapping Study of Security in Multi-Embedded-Agent Systems

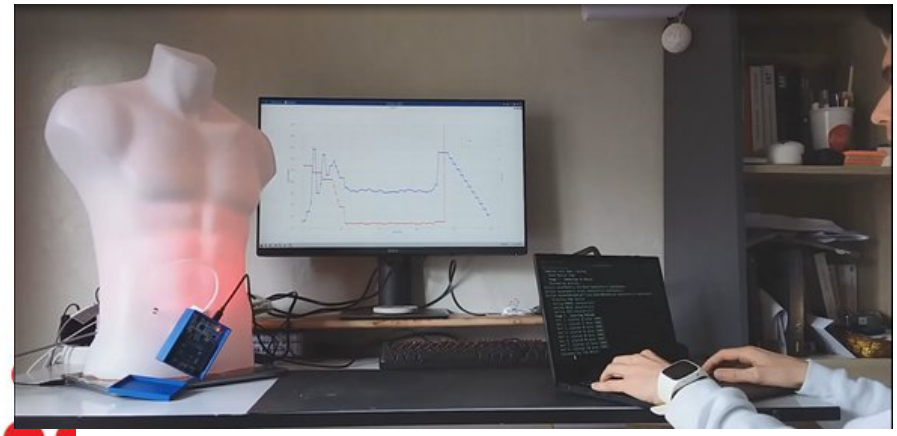
- Fault modeling for the analysis of vulnerabilities of embedded codes against fault injection
- Robustness evaluation of programmable components in critical systems

Main results and outcomes
Example of results

- SereneIoT project: Secured & EneRgy EfficieNt hEalth-care solutions using IoT technologies



Credit: C. Bresch PhD thesis



Connected secured insulin pump (Secpump)

Image credits: Esynov

- Main contributions:
  - Open-source life-critical medical device system workbench for security purposes
  - Critical IoT devices
  - **Global security (HW/SW) approach**

Main results and outcomes
Examples of ongoing projects

- **Hardware and software security of embedded systems**
  - Partners: CEA, CNRS, Inria, IMT, Grenoble INP – UGA
  - LCIS contribution: fault modeling and compilation-based security

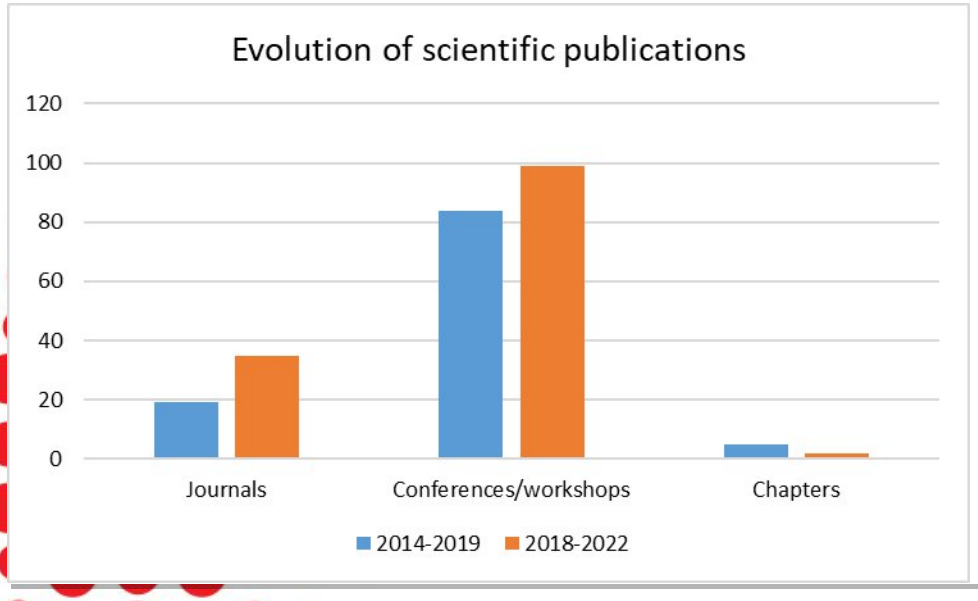
- **Hardware security of integrated systems**
  - ANR POP: Study of attacks on PUF
  - Partners: MSE, TIMA and LHC Labs
  - LCIS contribution: countermeasure design for PUF

- **Security of pervasive applications**
  - Project supported by the TRUST chair: heterogeneous connected devices and dynamic configurations
  - LCIS contribution: development of a middleware for the secure integration of heterogeneous devices

- **Security of mobile applications**
  - IDEX UGA CASTAV Project: vulnerability analysis of mobile applications depending on their context
  - Partner: LIG, LS2N
  - LCIS contribution: development of a tool for helping developers to develop secure mobile applications

Analysis of scientific production

	2018 Etpr : 9	2019 Etpr : 9	2020 Etpr : 9	2021 Etpr : 10	2022 Etpr : 10	<#ETPR>/ year
Journals	**8**	**5**	**6**	**7**	**6**	**0.68**
Conferences/ workshops	**14**	**21**	**13**	**13**	**17**	**1.66**
Chapters				**1**	**1**	



Increased scientific production

Analysis of scientific production

- Average **PhD** scientific publications:
  - 1 journal paper
  - 4 conference/workshop papers
  - 1 student is co-author of a patent
- International co-authors:
  - 48 publications **(1/3 of total publications)**
- International collaborations:
  - Europe: Austria, Czech Republic, Estonia, Germany, Greece, Italy, Sweden
  - America: Northern Arizona, California, South Carolina
  - Asia: Iran, Vietnam
  - North Africa: Algeria, Tunisia

Academic outreach

- Event organisation:
  - Organisation committees:
    - Conferences and workshops: 11 (JAIF, SHIFT, AITest, IE…)
    - Thematic schools: 2 (ARTISAN)
    - International contests and challenges: 6 (CSAW, AV Test)
  - Program committees: 12 (PLDI, Quatic, AITest, SHIFT…)
- Other achievements:
  - 2 Best paper awards (LOPSTR22, DSD22)
  - Penta innovation award (SereneIoT)
  - 8 invited talks
  - Average 6-month scientific stay abroad for 1 permanent and 2 PhD students

Analysis of strengths and weaknesses

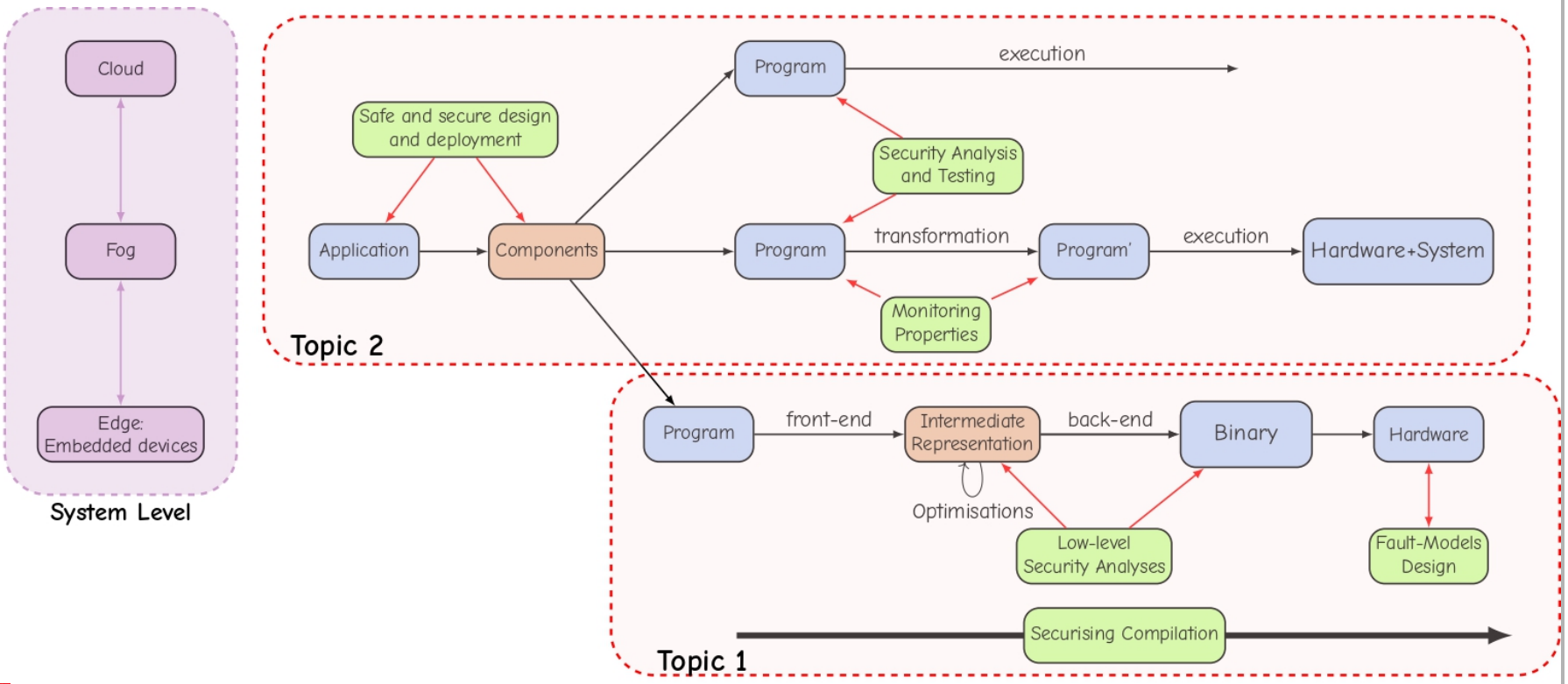
- Strengths:
  - National commitment (GDRs, French Computer Science Association…)
  - International visibility
  - Complementarity of expertise (thesis topics and supervising teams)

Perspectives

- Compilation-based safety and security: Language-based approach for generating secure binaries:
  - Fault model
  - Safety and security properties
  - Optimisation
- Security-by-design for autonomous and distributed systems:
  - Security analysis
  - Vulnerability modelling
  - Security testing

Strengthen the HS/SW complementarity for more safe and secure systems

Perspectives



Building blocks for a global safety and security hardware/software approach

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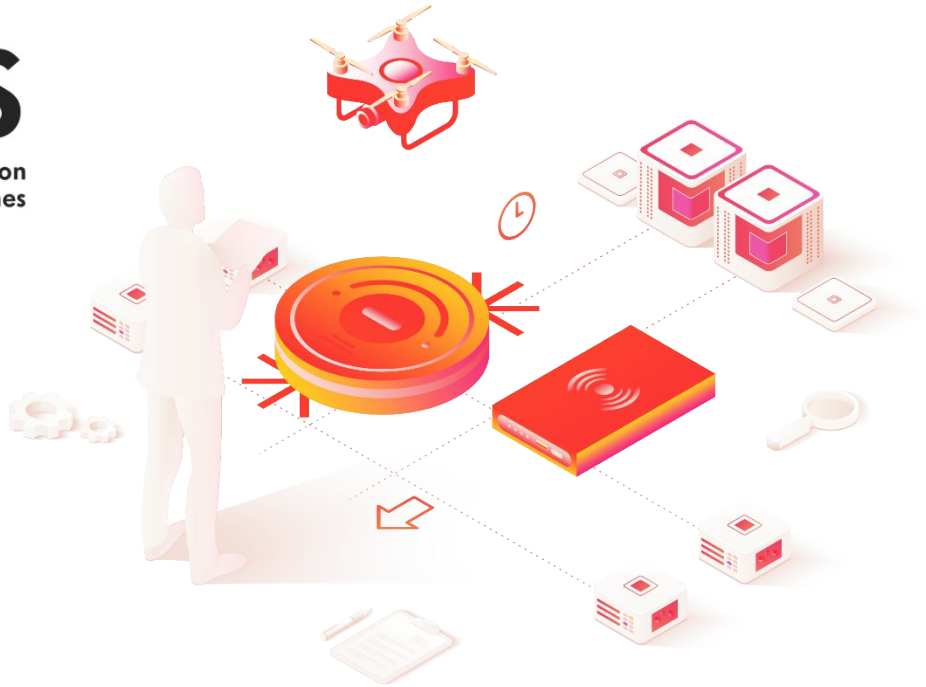


$(Co)^4Sys$

Coordination, Cooperation and Control of Complex Systems

Laurent.Lefevre@lcis.Grenoble-inp.fr

Staff Member

- **12 Permanents (4 Pr, 8 MCF)**
- Pr Laurent Lefèvre
- Pr Eduardo Mendes
- Pr Michel Occello
- Pr Jean-Paul Jamont
- MCF Ionela Prodan
- MCF Annabelle Mercie
- MCF Denis Genont-Catalot
- MCF Clément Ravieski
- MCF Simon Gay
- MCF Damien Genthial
- MCF François Suro
- MCF Louis Duvivier

- **Permanent staff: 12 academics**
  - ***Computer science*** *MAS (2 PU / 5 MC),* ***Network/electronics*** *(1 MC),* ***Control*** *(1 PU / 1 MC),* ***Modelling / numerics*** *(1 PU)*
  - *8 IUT - Valence, 3 Grenoble INP - Esisar , 1 DSDA*
  - *Heavy management/administrative* ***responsibilities and teaching duties****.*
- **PhD students**
  - Currently 12 PhD students, 7 working at LCIS / 11 theses defended in the 2018-2022 period (-1)

Scientific domain and research axes

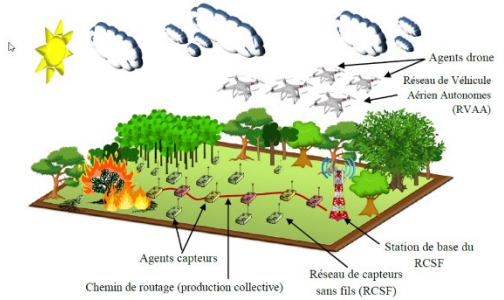
Distributed approaches for the analysis, modeling and control of complex systems

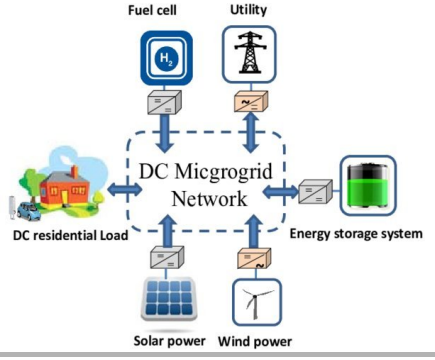
Research axes

- **Embedded Multi-Agent Systems (MAS)**
  - **Collective of Cyber-Physical Systems** (CCPS): drones, robots, WSAN, IoT, ...
  - **Non functional constraints** (energy, real-time, …), **robustness and resilience** (routing, safety/security, test/diagnosis), **environment models** and **trajectory planning/tracking**
    - drones, robots, Wireless Sensors/Actuators Networks (WSAN), Internet of Things (IoT)
- **Self-organization and emergence**
  - **collective output** emergence and identification, **cooperation** between MAS
  - **cohesion** and resilience
    - cyberattack collective detection, UAVs/Robots/Firemen coordination
- **Bio-inspired and social approaches for CPS**
  - **dynamical agents organizations** in MAS, **artificial emotions** and contagions
  - **developmental learning**, **incremental learning**
    - spatial localization, collective foraging, patrolling (fire spreads)
- **Complex Multiphysics systems**
  - **geometry** and **modelling** of physical systems (thermodynamics)
  - **analysis**, nonlinear **control** and constrained optimization-based control
    - grids/microgrids, water systems, biological/environmental systems

Main application fields

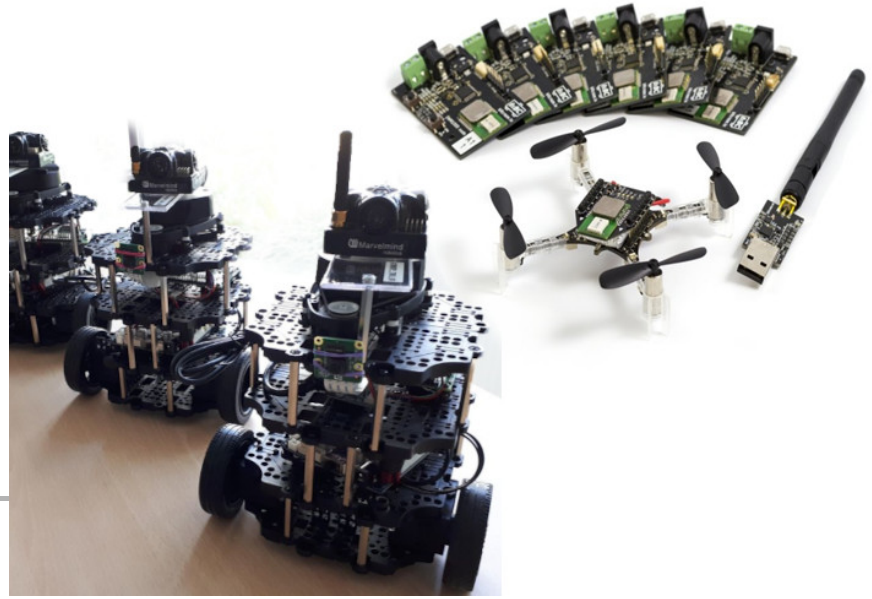
- **Collectives of cyber-physical objects**
  - drones (UAV, USV), mobile robots...
  - Internet of Things
  - Cyber resilience
  - Smart buildings
- **Supervision of "natural" environments**
  - Complex water systems
  - Landslides monitoring
  - Forest fire spreads
- **Infrastructure networks**
  - Grids/microgrids (DC, meshed, hierachical/distribute
  - Transport network (wasted/irrigation water, gas, hea
  - Distributed communication network (WSAN, CPL, ...)





Experimental platforms

- **CyberNetX:** localization, trajectory tracking/ planning and task assignment for CPS
  - UWB, and IR localization systems
  - Nano-drones (Crazyflies)
  - Mobile robots



Software platforms

- **MultiAgent Software-Hardware tool (MASH)**
  - A tool to design and to deploy **embedded MAS**
  - to test agent strategies with **realistic models of physical environments**
  - control the **functional deviation** between simulated agents and embedded implementation
- **Interoperable Software Framework (ISF, ongoing project)**
  - Lowering **interoperability** costs
  - Combine **physical robots**, **software agents** and **simulators**
  - Facilitate fruitful **collaborations** between team members
  - First step: evaluating frameworks (ROS, MQTT, DDS, ...) through PoC
- **Numerous simulation platforms supporting experimental work**
  - **Developmental learning** platform (EvoAgents, Vacuum-SG)
  - **Cyber-battle** simulator (MCAS)
  - Physical environment **cellular automata** simulator (CAMAS)
  - **Secure communication** with Multi-Agent Key Infrastructure (YAMMAS)

Main results (1/2)

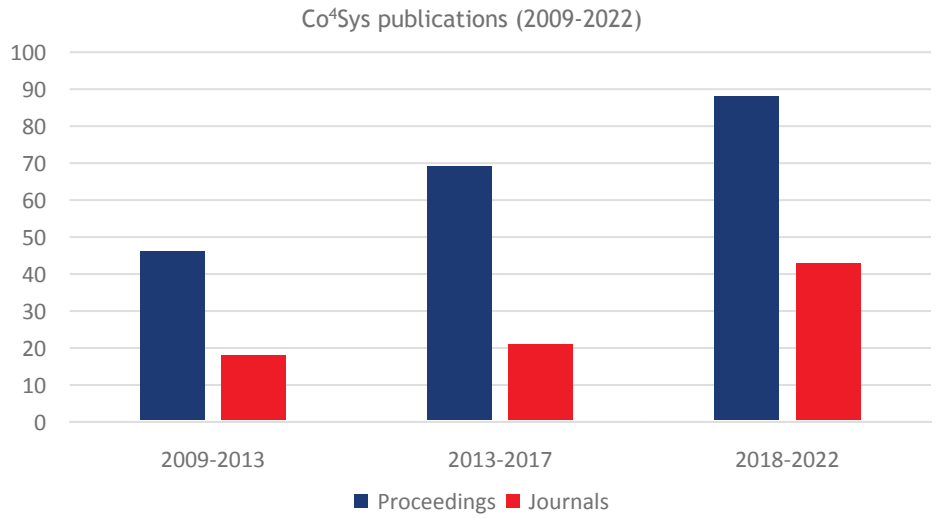
- **Embedded Multi-Agent Systems (MAS)**
  - **Models for organization and interactions** in collectives of embedded agents: MWAC (routing, resilience), CLASA (cybersecurity, resilience), MAKI (security key infratsructure)
  - **Test** of embedded multiagent systems
  - UAV/USV **Fleet control** and coordination
- **Self-organization and emergence**
  - **Interaction** between emerging systems: **exploitation of collective production**, **recognition of interaction** situations
  - **Multi-level** / **multi-scale** MAS architectures
  - Complex systems **analysis** and **identification** (information theory)

Main results (2/2)

- **Bio-inspired and social approaches for CPS**
  - Multi-agent **navigation**, learning **sparse topological environment model**
  - **CPS Resilience** utilizing cognitive functions of **emotions**
  - **Mobile sensors coordination** and social **organization** (distributed estimation)

- **Complex Multiphysics systems**
  - **Structure preserving discretization** (Multiphysics network / doctoral school)
  - (nonlinear) **constrained optimization-based** (distributed) **control** for (smart) grid/microgrid systems (national research group NMPC/microgrids)

Scientific output (publications)

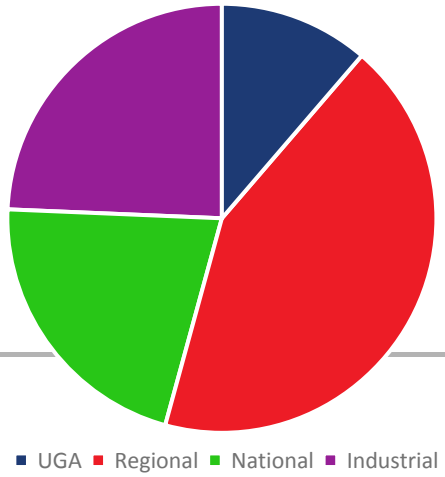


- **More journal papers (+100%) with relatively good visibility**
  - Both in **multi-agent systems** and **control**
  - **22xQ1 , 11xQ2, 6xQ3 and 3xQ4** journals (Scimago / SJR2 ranking) for 2018-2022
  - Still many papers in **specialized workshops / conferences**
- **Heterogeneity**
  - According to the scientific community (MAS vs. Control)
  - According to the people (theoretical vs. experimental subjects, service to the community, …)

Projects and funding

- **Local funding (Univ. Grenoble Alpes)**
  - **4 PhD theses** (+2 discharges from teaching duties)
  - 7 projects (5-30 k€) : thesis support and operating budget
- **Regional funding (AURA regions)**
  - 2 PhD theses
  - **2 projects** (resp. 131 and 191 k€) : PhD salary and thesis support
- **National funding (ANR, PHC, DGA)**
  - 5 PhD theses
  - **3 ANR projects** (36, 67, 164 k€) : thesis support and operating budget
  - 2 ongoing ANR projects : IMPACT (2022-2025) and MaestrIoT (2022-2025)
  - 2 PHC projects (France-Portugal, France-Great Britain) : travel and accommodation costs
- **Industrial funding**
  - **CIFRE / ANRT : 4 PhD theses**
  - Chaire TRUST : ½ PhD thesis
  - Chaire CybAir : 1 PhD thesis + 45 k€
  - Floralis / IESA : **7 contracts** (3,5-41 k€)

Co4Sys funding (2018-2022) : 752 k€



Academic outreach (2018-2022)

- **Workshop/conference organizations**
  - **9 invited sessions, 2 NOC, 8 IPC**
  - **IEEE SMC, IEEE CSS, IFAC, AFIA**
- **Responsibilities in science societies**
  - **GDR MACS, TC NMPC (**National research group in control, TC on nonlinear model predictive control)
  - **IEEE SMC France** (Systems, Man and Cybernetics)
  - **AICA IWG** (NATO task group on Autonomous Intelligent Cyber-defense Agents)
- **Invitations (research internships, invited lectures, graduate courses)**
  - **6 research visits** (Georgia Tech, HCM-UT, Univ. Porto, NTNU, UCL, Politehnica Bucarest)
  - **13 invited talks** (in workshops / conferences / seminars)
  - **4 invited lectures** (in international postgraduate courses)
- **Editorial activities**
  - **7 int rev, 1 int conf** (ECC, JPControl, Automatica, IEEE TAC, ROIA, …)
  - **2 books** (MAS), **1 special issue** (ROIA)
- **International academic network**
  - **Complex systems** (Universita di Firenze, Université de Genève)
  - **Drone control** (Universita di Udine, Univ. of Porto)
  - **Microgrid resilience and control** (UC London)
  - **USV navigation** (SINTEF/NTNU Trondheim)
  - **Multiphysics systems** (TU Munich, ITA Brazil, UC Louvain)
  - **Distributed cognitive systems** (CINVESTAV, Guadalajara)

Ongoing research and prospects

Cross-pollination between control and multi-agent research

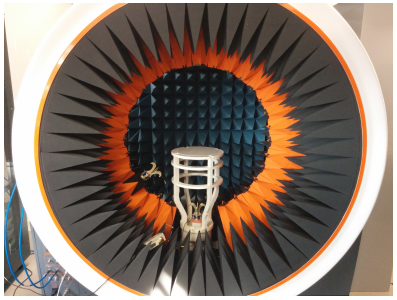
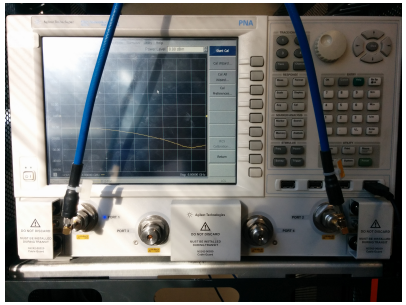
- **Distributed approaches for CPS**
  - **Learning for CPS** (developmental, reinforcement, bio-inspired SLAM)
  - **Resilience for distributed CPS** (cohesion, trust, test, diagnosis, and supervision)
  - **Heterogeneous CPS** (coordination/cooperation, auto-organization, social organizations)
  - **Complex multiphysics systems** (hierarchical/optimization-based control, process control, PHS)
- **Environment**
  - **MIAI** (Multidisciplinary Institute in Artificial Intelligence), **PERSYVAL-lab** (Pervasive Systems and Algorithms), **Cybersecurity Institute**
  - **GDR MACS**, **AFIA**, **IEEE SMC**, **IFAC**
  - **CyberNetX platform** : drones, robots, wireless sensors/actuators networks (WSAN)
  - **ISF** (Interoperable Software Framework)

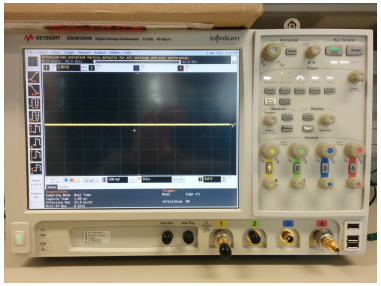
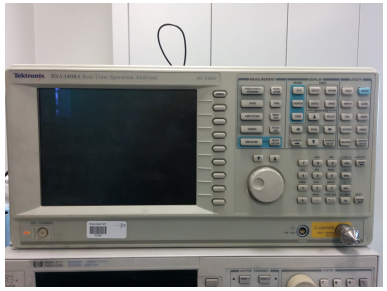
Platform

Available platforms

Experimental Platform

- Plateforme RF pour la mesure et la caractérisation RF en conduit et sans fil
- Bande de fréquence de 0 Hz à 77 GHz
- 4 VNA, 2 DSO, 2 VSG, 1 SA, 1 CA

Resources

esynov
by Grenoble INP - Esisar, UGA

Four experimental platforms

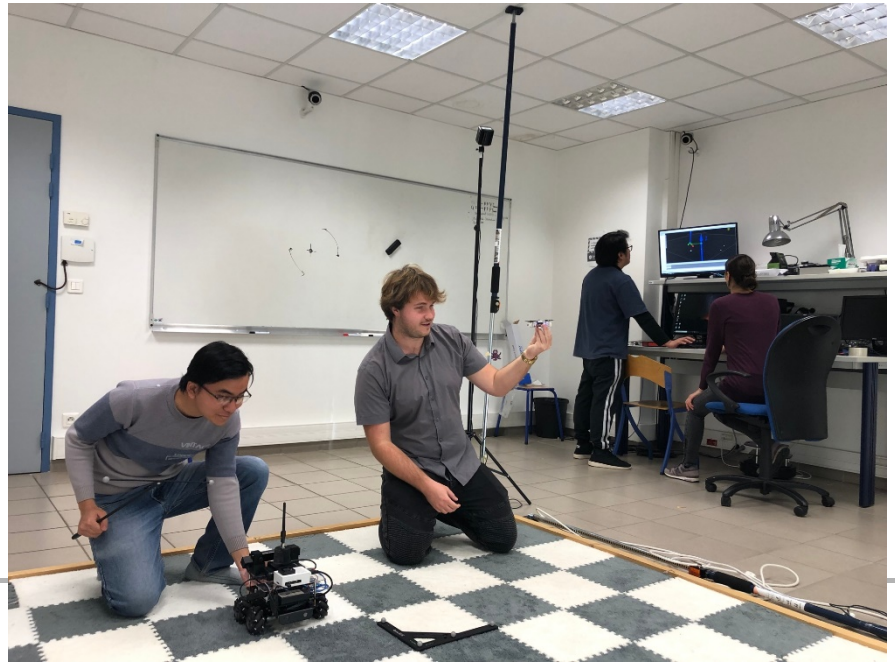
- « RF Measurements » (ORSYS)
- « Cybersecurity of embedded systems » (Sacco)
- « CPS localisation and task assignment» (CyberNetX)

Resources

Four experimental platforms

- « RF Measurements » (ORSYS)
- « Cybersecurity of embedded systems » (Sacco)
- « CPS localisation and task assignment» (CyberNetX)
- « Sensor/actuator networks » (Sycadi)

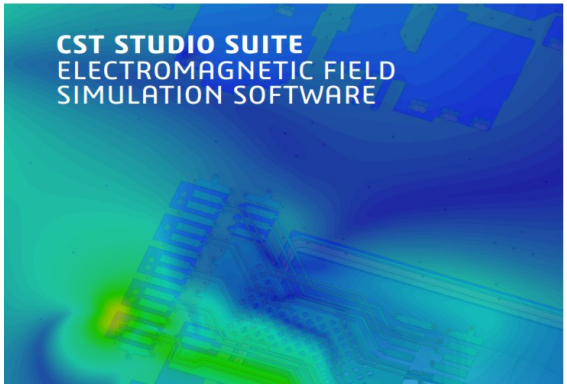
esynov
by Grenoble INP - Esisar, UGA

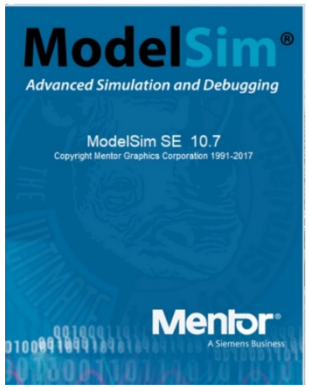


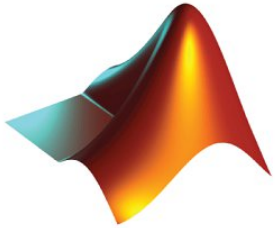
Resources

Workstations

- ~ 100 personal computers
- **7 scientific servers**







MATLAB

- **Co-management of services of the DSI Esisar** (antivirus, backup, etc.)
- **Technical service (3 members)**
  - Research engineer
  - Technician
  - Engineer
- **Additional social space in building D** (from 2022 due to LCIS moving)

Scientific animation

Scientific animation

Scientific animation

Agenda of 2022/2023 activities

Activity	Date
Fête de la science/Science Fair	October (yearly)
Kaléidoscope Week Grenoble INP - UGA	October 2022 (yearly)
LCIS Scientific Day	March/May (yearly)
LCIS Scientific Council/HCERES	May 2023 (every 2.5 years)
PhD Day	June (yearly)
SpeechPizza Talks	Once per month
Seminars	Regularly (usually once per/month)
Other activities (Summer schools, Workshops, GDR MACS, CSAW)	

Scientific animation

Foster research to a larger public

- **Science Fair (yearly session)**
  - Experimental platforms and prototypes
  - Number of visitors: ~ 4900

- **K… session)**
  …GA
  …is within LCIS

GRENOBLE INP UGA — LCIS Laboratoire de Conception et d'Intégration des Systèmes — UGA Université Grenoble Alpes

Présentation et visite du LCIS

Semaine Kaléidoscope Grenoble INP - UGA 2022
Jeudi 27 Octobre 2022
lcis.grenoble-inp.fr

Le LCIS est un laboratoire de recherche public de Grenoble INP et de l'Univ. Grenoble Alpes sur le campus UGA Valence. Il rassemble plus de 60 chercheurs en informatique, électronique et automatique autour des systèmes embarqués et communicants. Ses thématiques de recherche concernent la sûreté et la sécurité des systèmes embarqués et distribués, la modélisation, l'analyse et la supervision des systèmes complexes ouverts et les systèmes radiofréquence sans fil communicants. Ses domaines d'application sont variés : internet des objets, systèmes cyber-physiques, environnements connectés naturels ou artificiels, RFID.

Session du matin en salle A042
9h00 - 9h30 : Présentation générale des activités de recherche du LCIS et de ses équipes (I. Prodan, dir. adj. LCIS).
9h30 - 10h10 : Visite de la plateforme Esisarium (planification des mouvements et commande des systèmes cyber-physiques).
10h10 - 10h50 : Visite de la plateforme RF (mesures d'antennes, caractérisation des matériaux et analyse spectrale).
10h50 - 11h30 : Visite de la plateforme SACCO (injection de fautes, attaques par canaux cachés et contre-mesure).

Session de l'après-midi en salle D030
13h00 - 14h30 : Présentation générale des activités de recherche du LCIS et de ses équipes (V. Beroulle, dir. LCIS).
L'équipe CO4SYS : Coordination, coopération, contrôle des systèmes complexes (L. Lefèvre)
L'équipe CTSYS : Sûreté et sécurité des systèmes embarqués et distribués (O. Aktouf)
L'équipe ORSYS : Systèmes RF sans fil communicants (N. Barbot)
Qu'est ce qu'une thèse : Différents aspects seront abordés, par ex., pourquoi faire une thèse, comment est elle financée, quels débouchés après la thèse, la vie du chercheur dans une équipe de recherche (en présence de différents PhDs du LCIS).
14h45 - 16h00 : Visites de certains plateformes du LCIS (Esisarium, RF, SACCO).

Clôture de la journée et échanges avec les chercheurs du LCIS autour d'un café

Inscription en suivant le lien : lien pour l'inscription. Pour plus d'informations concernant l'organisation de la Journée, vous pouvez contacter I. Prodan, dir. adj. LCIS, ionela.prodan@lcis.grenoble-inp.fr.

50 rue Barthélémy de Laffemas, 26000, Valence.

Qu'est-ce que la recherche scientifique ?

Scientific animation

Encourage research interactions

- **LCIS Scientific Day (yearly session)**
  - External invitees talks (at least 3)
  - Poster Sessions (internal and external PhDs)
  - Industrial invitees
  - Number of participants each year: 50 to 80
- **LCIS PhD Day (yearly session)**
  - PhDs and intern students presentations
  - Social activities
  - Number of participants: 50 to 60
- **SpeechPizza talks (monthly)**
  - During lunch break
  - Dedicated entirely to the PhDs interactions
- **Seminars (regularly/monthly)**
  - Invited professors or PhDs/postdocs
- **Feedback coffee**: short sessions dedicated to call for proje

GRENOBLE INP UGA — LCIS Laboratoire de Conception et d'Intégration des Systèmes — UGA Université Grenoble Alpes

SpeechPizza

Thursday, 20 April 2023
12:00 - 13:00, D011
lcis.grenoble-inp.fr

Sébastien Michelland sebastien.michelland@lcis.grenoble-inp.fr
(1st year PhD candidate within CTSYS team, supervised by L. Gonnord and C. Deleuze)

Title: The time and place for language semantics
Abstract: Semantics is the art of writing down rigorously what programs do. But why, or better still, when is that important? Why is source code not already a self-description of the program? What does this weird symbol "" even mean?! This talk will tackle semantics from a non-specialist's perspective to provide an intuition of when semantics can help and what to do when encountering them in the wild.

Huu Thinh Do huu-thinh.do@lcis.grenoble-inp.fr
(2nd year PhD candidate within CO4SYS team, supervised by I. Prodan)

Title: Indoor experimental validation of MPC-based trajectory tracking for a quadcopter
Abstract: Differential flatness has been used to provide diffeomorphic transformations for non-linear dynamics to become a linear controllable system. This greatly simplifies the control synthesis since in the flat output space, the dynamics appear in canonical form (as chains of integrators). The caveat is that mapping from the original to the flat output space often leads to nonlinear constraints. In particular, the alteration of the feasible input set greatly hinders the subsequent calculations. In here, we particularize the problem for the case of the quadcopter dynamics and investigate the deformed input constraint set. An optimization-based procedure will achieve a non-conservative, linear, inner-approximation of the non-convex, flat-output derived, input constraints. Consequently, a receding horizon problem (linear in the flat output space) is easily solved and, via the inverse flat mapping, provides a feasible input to the original, nonlinear, dynamics. Experimental validation and comparisons confirm the benefits of the proposed approach and show promise for other class of flat systems (the video is available at: https://youtu.be/1a1K6R6__3s).

The science behind pizza!

The volunteers for giving a short talk the next editions should not hesitate to contact asap ionela.prodan@lcis.grenoble-inp.fr.

Scientific animation

National/international scientific activities

- **Workshops**
  - JAIF 2022 (thematic day on fault injection attacks)
  - More than 150 participants
- **Summer schools**
  - ARTISAN 2022 (~ 30 students)
  - School for Young Researchers in Programming (~ 50 students expected)
- **CSAW (in collaboration with Esisar since 2017)**
  - International contest in Cybersecurity
  - More than 70 contestants
- **GDR MACS**
  - GDR Ondes (~20 participants)
  - GDR CPNL, expected in Nov. 2023

Self-assessment

The number of publications is steadily increasing

- two consecutive periods are analyzed (2014-2018 and 2018-2022)

	2014 –2018	
	Total	Mean /year
Journals	**102**	20.4
(Inter)national conferences	292	58.4
Books	21	4.2
	415	**83**

Outreach: Highlights

- **Various prices and recognitions**
  - Grand Prix de l'Electronique du Général Ferrié 2021
  - Prix Espoir IMT – Académie des sciences 2020
  - IEEE MTT-S 2019 Outstanding Young Engineer Award
  - PENTA Award for innovation for the European project SERENE-IoT 2021
- **Strengthening of collaborative and transfer projects on various topics**
  - An ERC project, SereneIoT European project
  - Chaire Trust, R&D Booster (Anaconda, Detoxio-S, Wyzeflow), Linksium (EDA4SEC, HRFID)
- **Better visibility and involvement in scientific and technical committees**
  - (Inter)national events organisation (CSAW, JAIF, AITEST, SHIFT, WPW…)
  - Editorial, program and technical commitees, Conference vice-chair, WIC chair

Outreach: Highlights

- **Active involvement in various committes/clusters within UGA**
  - Deputy director MSTIC cluster
  - Labex Microelectronics council
  - Persyval, MIAI, FMNT axis
  - GINP IRT PULSE
  - Chair Trust council
- **Other responsabilities within UGA**
  - Director HR Grenoble INP, deputy director HR EUT UGA,
  - Director of Esisar and site Esisar, council Esisar, specialties (MISTRE, IR&C, EIS) and departments responsabilities at Esisar and IUT, Campus UGA Valence council

Attractiveness through scientific dissemination

Involvement in various technical committees

- **National level:** Chair of GdR MACS (CPNL), organisation of GdR Ondes, GPL, National day of young researchers in informatics, thematic schools (Persyval, young researcher), organisation of workshops and conferences (JAIF)

- **International level**: contests and conferences (CSAW, AITEST, SHIFT, WPW, URSI-GASS, IEEE-RFIDTA, IEEE-MTTs, IEEE-Aps EuCAP), editorial commitees (IEEE ECC, IFAC CAO…), summer schools organisation, scientific organisations: IEEE, vice-chair URSI, program committee at different conferences, best paper award jury (SIF'20'21'22, IFAC CAO'22, ECC'23)

- **Various invited researchers and Professors within the lab**
- **Various talks and seminars outside the lab:** SOSE 2022, Georgia Tech, USA, IFAC CAO'22, HCMUT, Vietnam, UCL London, Univ. of Porto, MASCARA 2020..

Attractiveness through quality of staff reception

- **Interns, PhDs and post-docs**
  - prioritize their participation to conferences, scientific courses, mobility projects and summer schools
  - organize yearly PhD Day and monthly Seminars (SpeechPizza) dedicated to young researchers interactions
  - thesis monitoring committees proposed by EEATS and MSTII

- **New EC arrivals**
  - discharge of 62 hours of teaching
  - 5k euros funding for functioning
  - Internship funding proposals, specific research support
- **From 2018 to 2022, LCIS organized 5 EC recruitments** (only one recruitment proved unsuccessful)

New challenges and projects

- **Strengthen the support of young researchers for local/regional projects applications** (e.g., IRGA, PersyvalLab, FMNT, EEATS, MSTII, Pack Ambition Recherche, ANR…)
- **Develop inter-teams projects through the existing platforms** (e.g., CyberNetX can cover various application domains in Robotics, Multi-agent Systems, Sensing and localisation, Cybersecurity, etc.)
- **Develop industrial Chair(s) in collaboration with Esisar which can cover the various research themes of LCIS** (e.g., the scientific day in March 2023 had a session dedicated to this reflection)

New challenges and projects

- **Integration/management of new national and European projects which tackle current societal challenges**
  - e.g., PEPR Cybersécurité, PEPR hydrogène décarboné, PEPR exploratoire O2R (robotique), PEPR d'accélération Électronique

  - e.g., MSCA Doctoral Networks, industry and academic collaborations on topics as security and safety in HW/SW components, IoT security, energy management, precision agriculture etc.

- **Encourage mobility of our EC and invited researcher**